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KAPASA MAKASA UNIVERSITY

CYBER SECURITY

Fraud Prevention & Detection Technologies (CYS339)

EXPECTED QUESTIONS

Questions by James Soko

UNIT 1 – DATA ANALYSIS

1. What is computer data? [2 Marks]
2. What do you understand by the term analysis? [2 Marks]
3. What is data analysis? [2 Marks]
4. Define the term fraud. [2 Marks]
5. Define the term fraud in relation to cybersecurity. [2 Marks]
6. Give at least four (4) examples of a cybersecurity fraud. [4 Marks]
7. What are detection technologies? [2 Marks]
8. List at least four (4) detection technologies. [4 Marks]

TOTAL: [20 Marks]

9. List and explain the three reasons why data analysis is a must. [3 Marks]
10. Data analysis can be done by both auditing and investigation. Understanding these two concepts is essential for you to safe guard your digital asserts.
    a) Mostly when we are dealing with finances we use auditing, even in cybersecurity an IT security audit is conducted periodically.
        i. What is a cybersecurity audit? [2 Marks]
        ii. What is the purpose of a cybersecurity audit? [2 Marks]
        iii. What is the scope of a cybersecurity audit? [2 Marks]
        iv. List at least two (2) tools used in cybersecurity auditing. [2 Marks]
        v. List at least three (3) auditing techniques that can be used in cybersecurity. [3 Marks]

    b) The most common method used in cybersecurity is a forensic investigation when an incident occurs.
        i. What do you understand by the term investigation? [2 Marks]
        ii. What is the purpose of conducting an investigation? [2 Marks]
        iii. What is the scope of a cybersecurity investigation? [2 Marks]

TOTAL: [20 Marks]

        iv. List at least three (3) tools used for conducting an investigation. [3 Marks]
        v. List at least three (3) techniques used in a cybersecurity investigation. [3 Marks]
11. List at least five (5) things that data analysis can do. [5 Marks]
12. List and explain two (2) things what data analysis cannot do. [4 Marks]

TOTAL: [15 Marks]

UNIT 2 – COMMON ANALYSIS PROCEDURES & FUNCTIONS

COMMON ANALYSIS PROCEDURES

Financial Ratio

1. What do you understand by the term financial ratio? [2 Marks]
2. In Cybersecurity financial ratio helps the CISO to assess 4 major factors namely – [4 Marks]
3. What are the 3 advantages of financial ratio? [3 Marks]
4. What are the 3 limitations of financial ratio? [3 Marks]
5. A company spends $50,000 annually on cyber security measures. The total financial loss due to cyber-attacks before implementing the measures was $200,000 and after implementation the loss reduced to $80,000 Dollars. Calculate the Return on Investment. [8 Marks]

TOTAL: [20 Marks]

Variance & Percentage

6. What do you understand by the term variance in relation to fraud detection? [2 Marks]
7. List and explain the 2 types of variances. [2 Marks]
8. A company records an expected daily login count of 5000 for each employee. On a particular day, the actual recorded log in count was 7500.
   i. Calculate the variance. [2 Marks]
   ii. What is the interpretation of the variance result? [1 Mark]
9. Explain at least two advantages of variance analysis in fraud detection. [2 Marks]
10. Give 1 disadvantage of variance in fraud detection. [1 Mark]
11. What do you understand by the term percentile analysis in fraud detection? [2 Marks]
12. List two major types of percentages and give the basic formulars. [2 Marks]
13. A cybersecurity team analyzes 500 phishing attempts in a month. Out of these, 35 were successful. Calculate the attack success rate percentage. [2 Marks]
14. Explain at least two advantages of percentile analysis in fraud detection. [2 Marks]
15. Explain 3 limitations of percentile analysis in fraud detection. [2 Marks]

TOTAL: [20 Marks]

Aging & Average (Mean)

16. What do you understand by the term aging? Give 3 scenarios how it can be applied in fraud detection. [4 Marks]
17. A company tracks unpatched vulnerabilities overtime. The data is as follows:

Age (Days)	Unpatched Vulnerabilities
0 – 30	25
31 – 60	20
61 – 90	10
91+	5

    i. What percentage of vulnerabilities are older than 60 days? [2 Marks]

18. Give 1 advantage of aging in fraud detection. [1 Mark]
19. List 2 major disadvantages of aging in fraud detection. [2 marks]
20. What is an average used for in fraud detection? [1 Mark]
21. A security operations center measures the mean time to detect cyber threats over 5 incidents. Detection Times (hours): [4,5,3, 6,7]. Calculate the mean time. [6 Marks]
22. Give 2 advantages of mean in fraud detection. [2 Marks]
23. List 2 major disadvantages of mean in fraud detection. [2 Marks]

TOTAL: [20 Marks]

FUNCTIONS

1. What is the main purpose of filtering? [2 Marks]
2. Give at least two (2) advantages and disadvantages of filtering. [2 Marks]
3. What is the main purpose of sorting data? [2 Marks]
4. Give at least one (1) advantage and disadvantage of sorting. [2 Marks]
5. Join and append are two (2) prevalent data analysis functions. Highlight the key similarities and differences between these two functions. [3 Marks]
6. Give at least two (2) advantages and disadvantages of join and append. [4 Marks]
7. What is the main use of a pivot in analysis? [2 Marks]
8. Give at least two (2) advantages and disadvantages of pivoting. [2 Marks]
9. What is trend analysis? [2 Marks]
10. Give at least one (1) advantage and disadvantage of trend analysis. [2 Marks]
11. What is regression? Also, least its two (2) common types. [3 Marks]

12. Give at least two (2) advantages and limitations of regression. [2 Marks]

13. A company's revenue depends on the amount spent on advertising, the relationship between sales and the advertising spent is modeled by the following regression equation. If a company spends $2000 on advertising, predict the expected sales revenue using the equation:

Sales = $500 + (10 * Advertising)

14. What is the main purpose of grouping in data analysis? [2 Marks]

15. What are the key characteristics of grouping in data analysis? [4 Marks]

16. List and explain the two types of grouping. [4 Marks]

17. Give at least two (2) advantages and disadvantages of grouping. [2 Marks]

18. What is summarization? [2 Marks]

19. List and explain the three (3) summarization techniques. [3 Marks]

20. List at least two (2) types of charts you can use to summarize data. [2 Marks]

21. List at least two (2) advantages and disadvantages of summarization. [2 Marks]

22. What is stratification? Also, list and explain the four (4) characteristics of stratification. [5 Marks]

23. List at least two (2) advantages and disadvantages of stratification. [2 Marks]

24. What is the main use of a VLOOKUP function? [2 Marks]

25. Provide a basic syntax of a VLOOKUP excel function. [2 Marks]

26. How does a VLOOKUP excel function work? [2 Marks]

27. Give at least two (2) basic usage of a VLOOKUP excel function. [2 Marks]

28. What are some of the limitations of a VLOOKUP function? [2 Marks]

29. What are the alternatives of a VLOOKUP function? [3 Marks]

UNIT 3 – DATA ANALYSIS

Data analysis techniques in fraud prevention and detection systems come in two types which are quantitative and qualitative. Understanding these two data analysis types is essential for cybersecurity experts.

1. Quantitative data analysis must be critically understood because it has a lot of types that are necessary for analysis.
    a) What is quantitative data analysis? [2 Marks]
    b) List and explain the eight (8) types of quantitative data analysis by highlighting their cybersecurity applications and tools (Hint: define, cybersecurity applications and tools). [16 Marks]
    c) Despite quantitative data analysis having many major types, it also has other four (4) types. List and explain its other types by also highlighting their tools, strengths and limitations. [7 Marks]

    TOTAL: [25 Marks]

2. Qualitative data analysis is necessary when cybersecurity experts want to gather data for threat intelligence so has to increase its security posture.
    a) What is qualitative data analysis? [2 Marks]
    b) List and explain the two (2) major qualitative data analysis techniques by highlighting their tools, strengths and limitations. [4 Marks]
3. Highlight the four (4) major deference's between qualitative and quantitative data analysis using the following factors: data type, analysis focus, tools and outcome. [4 Marks]

    TOTAL: [10 Marks]

UNIT 4 – GAP ANALYSIS

1. What is gap analysis? [2 Marks]
2. How can gap analysis help an organization against cyberthreats? [3 Marks]
3. Highlight the steps involved in conducting gap analysis for cybersecurity fraud prevention. [5 Marks]
4. How can you use gap analysis in fraud detection? [2 Marks]

TOTAL: [12 Marks]

UNIT 6 – PERFORMING CONTEXUAL AND BEHAVIOR QUALITATIVE ANALYSIS

1. List and the three (3) major concepts used in contextual and behavioral analysis. [3 Marks]
2. Most fraudulent activities are conducted online, list and explain the three (3) online presence and social media analysis techniques. [3 Marks]
3. Forensics is commonly used to investigate digital evidence. Explain the digital forensic process that can be used to detect fraud. [5 Marks]
4. List and explain the three (3) common forensic techniques you can use in fraud detection. [3 Marks]
5. List and explain the three (3) challenges faced in conducting a fraud investigation. [3 Marks]

TOTAL: [17 Marks]

UNIT 7 – OLAF APPROACH IN DATA ANALYSIS

1. What do the initials OLAF stand for? [2 Marks]
2. What is the role of OLAF in data analysis? [2 Marks]
3. A company experienced fraud in the past years, as a Cybersecurity analyst, you've been employed to assess these fraudulent activities on an organizations IT infrastructure. Using the OLAF approach, highlight the key steps you can take to analyze and fraud. [8 Marks]
4. List at least two (2) challenges associated with the OLAF approach. [2 Marks]
5. What is the difference between miscarriage of justice and impurity of findings? Also highlight their causes and preventives measures. [6 Marks]

TOTAL: [20 Marks]